LIBRARY MANAGEMENT SYSTEM

Database Design and Normalization Project

Database platform: PostgreSQL 17

1. Project Overview

The Library Management System database stores information about library members, books, book categories, authors, and book loans. It uses relational tables and primary/foreign key constraints to keep data organized and maintain relationships between records.

2. Objectives

- Design a relational database with at least five related tables.
- Use primary keys and foreign keys to identify records and maintain referential integrity.
- Model the many-to-many relationship between books and authors.
- Apply normalization principles, including Third Normal Form (3NF).
- Create and test the schema using PostgreSQL.

3. Database Tables

Table	Purpose	Key columns
Members	Stores library member details.	member_id (PK), email (unique)
Categories	Stores book categories.	category_id (PK), category_name (unique)
Books	Stores book details and category.	book_id (PK), category_id (FK), isbn (unique)
Authors	Stores author details.	author_id (PK)
Book_Authors	Connects books and authors.	Composite PK (book_id, author_id); both are FKs
Loans	Tracks borrowed books and due/return dates.	loan_id (PK), member_id (FK), book_id (FK)

4. Relationships

- Categories to Books: one category can contain many books; each book belongs to one category.
- Books to Authors: many-to-many, implemented by the Book_Authors junction table.
- Members to Loans: one member can have many loan records.
- Books to Loans: one book can appear in many loan records over time.

The ER diagram exported from dbdiagram.io should be inserted or attached alongside this report.

5. Normalization and 3NF

First Normal Form (1NF): Each table has rows and columns with single values in each field. Author information for a book is not stored as a comma-separated list; each book-author pair is stored as a separate row in Book_Authors.

Second Normal Form (2NF): Tables with a single-column primary key have attributes that depend on that key. In Book_Authors, which has the composite key (book_id, author_id), there are no additional non-key attributes, so there are no partial dependencies.

Third Normal Form (3NF): Non-key attributes depend on the key, the whole key, and nothing but the key. Category names are stored in Categories rather than repeated in Books, and author names are stored in Authors rather than repeated in Books. This reduces duplication and helps prevent update, insertion, and deletion anomalies.

6. Primary Keys, Foreign Keys, and Indexes

Primary keys uniquely identify rows. Foreign keys connect related tables and prevent references to missing parent records. UNIQUE constraints are used for member email, category name, and ISBN where provided. Explicit indexes support common lookups on Loans.member_id, Loans.book_id, and Books.category_id. PostgreSQL also creates indexes for primary-key and unique constraints.

7. Validation Rules

The Loans table checks that due_date is not earlier than borrow_date and that a non-null return_date is not earlier than borrow_date. The borrow_date column defaults to the current date.

8. Testing

The schema and sample records were created in PostgreSQL. Use the following query to verify that loan records correctly connect members and books:

```
SELECT m.name AS member_name, b.title AS book_title, l.borrow_date, l.due_date, l.return_date
FROM Loans l
JOIN Members m ON l.member_id = m.member_id
JOIN Books b ON l.book_id = b.book_id;
```

A successful result displays the member name, book title, borrowing date, due date, and return date for each loan.

9. Sample Data

The sample dataset includes categories Fiction, Science, and Technology; authors R. Sharma and A. Verma; members Rahul Kumar and Priya Singh; and books The Science World and Learning SQL. A loan record is added for member 1 and book 1.

10. Deliverables

- This project report.
- ER diagram PDF exported from dbdiagram.io.
- SQL script containing table definitions, constraints, indexes, sample data, and test queries.

Conclusion

The Library Management System database contains six related tables and demonstrates relational modeling, primary and foreign keys, a many-to-many junction table, indexing, validation constraints, sample data, and the principles of Third Normal Form.